

BY 613: Metagenomics, and Next Generation Sequencing Technologies

Course Code	BY 613
Course Name	Metagenomics, and Next Generation Sequencing Technologies
L-T-P-C	3-0-0-3
Prerequisites	BY 606 or consent from the faculty
Intended for	B. Tech. M. Tech., Ph.D.
Distribution	Core for M. Tech. Biotechnology, elective for others
Approval	11th Senate (Oct 2016)

Course Contents

- **Next-generation sequencing technologies**

Introduction to different sequencing technologies including Capillary based (Sanger), pyrosequencing (454), Illumina, Solid, and others. Differences in data generated by these methods such as read length, throughput, read qualities, etc. Tools and resources that are available to analyze the generated data from different sequencing methods. Types of sequencing data that can be produced from different sequencing methods, such as genomic, metagenomic, amplicon, exome, transcriptome, targeted sequencing, CHIP-seq, 16S, etc.[22 Lectures]

- **Metagenomics**

Topics are Metagenomics-specific issues like calling taxa, functional annotation, metagenome assembly and comparative metagenomics. This course will help participants answering questions like:[20 Lectures]

- What are possible approaches of metagenomics?
- How to monitor and predict environmental conditions and change?
- Biological insights, phylogenetic diversity.
- Examining genes/operons for enzyme and natural products.

Many hurdles arise in the analysis and assembly of environmental genomics records, among which short sequence reads, high species complexity in samples and the availability of specialized software for microbial genomics analysis.

Among the many topics in metagenomics addressed in this course are: experimental approaches; sequencing technologies; platforms and platform specific issues; methodologies; Sample preparation; QC-reports & quality controls; mapping sequence reads; taxonomic annotation; functional annotation & function determination; unknown gene prediction; comparative metagenomics; amplicon sequencing; shotgun metatranscriptomics; calling taxa; metagenome assembly; evaluation; visualization and reporting of results; tools & algorithms; 16S profiling; databases; mapability; k-mer profiling; cross assembly; QIIME; MG-RAST; iPath.

References

1. Victor Kunin et al, A Bioinformatician's Guide to Metagenomics, **MICROBIOLOGY AND MOLECULAR BIOLOGY REVIEWS**, Dec. 2008, p. 557–578.
2. Next-generation sequencing data interpretation: enhancing reproducibility and accessibility, **Anton Nekrutenko & James Taylor Nature Reviews Genetics**, Volume 13, No 9, 667, September 2012 .

Articles

Other Latest research articles will be advised related to the topic being taught from time to time.